

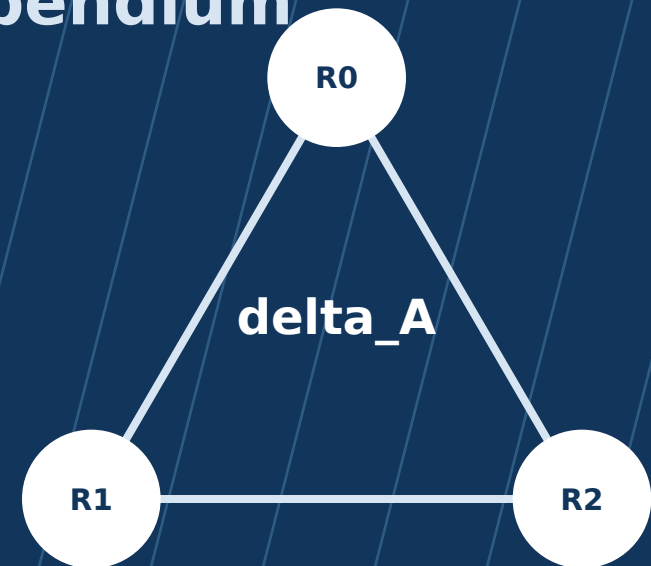
Universal Calculus

Kleban Operator Diagram Compendium

Diagrammatic translation of branched equalizer geometry

Primitive terminal equations

```
K_{\theta,\beta} = \delta_A \circ R^{\beta}_{\theta}
\text{Real}^{\beta}_0 = \ker K_{\theta,\beta}
\text{Real}^{\beta}_{\epsilon} = K^{-1}_{\theta,\beta}(B_{\epsilon}(0))
DK = \delta_A \circ dR^{\beta}_{\theta}
```



K = cyclic compatibility defect

Navigation

Primitive operator

01

Layer stack

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Stable comparison

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Global phase and branches

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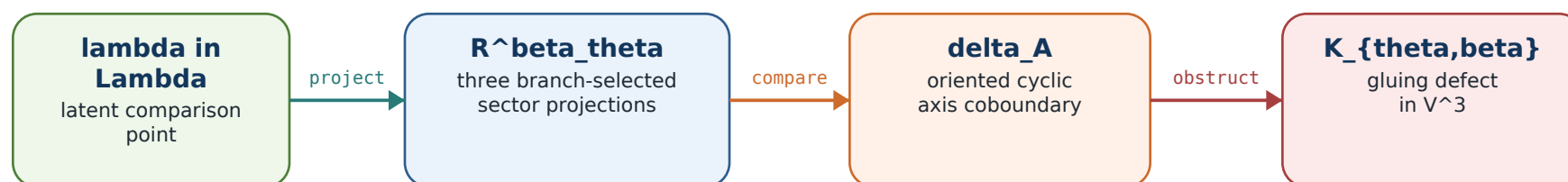
Unknown unknowns

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Terminal checklist

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Terminal semantic object



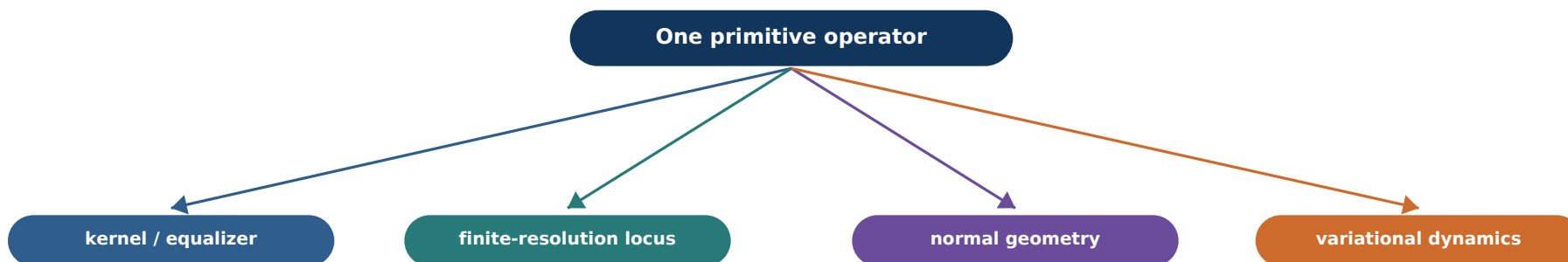
Definition

$K_{\{\theta,\beta\}} := \delta_A \circ R^{\beta,\theta}$
 $\delta_A(X_0, X_1, X_2) =$
 $(X_0 - X_1, X_1 - X_2, X_2 - X_0)$

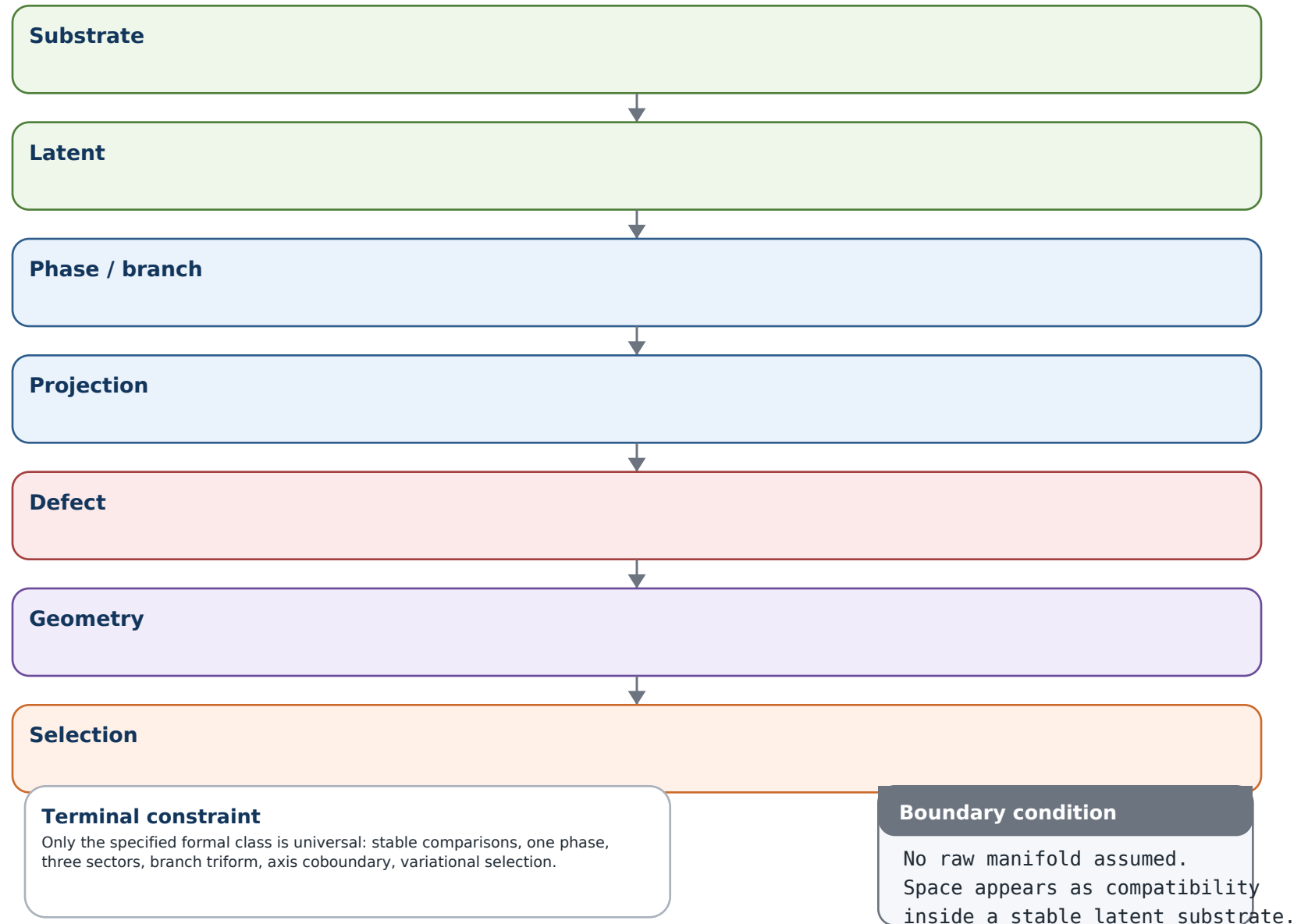
Generated interpretations

$K = 0$	-> exact agreement
$ K \leq \epsilon$	-> finite agreement
dK	-> infinitesimal defect
$ K ^2$	-> dynamical cost

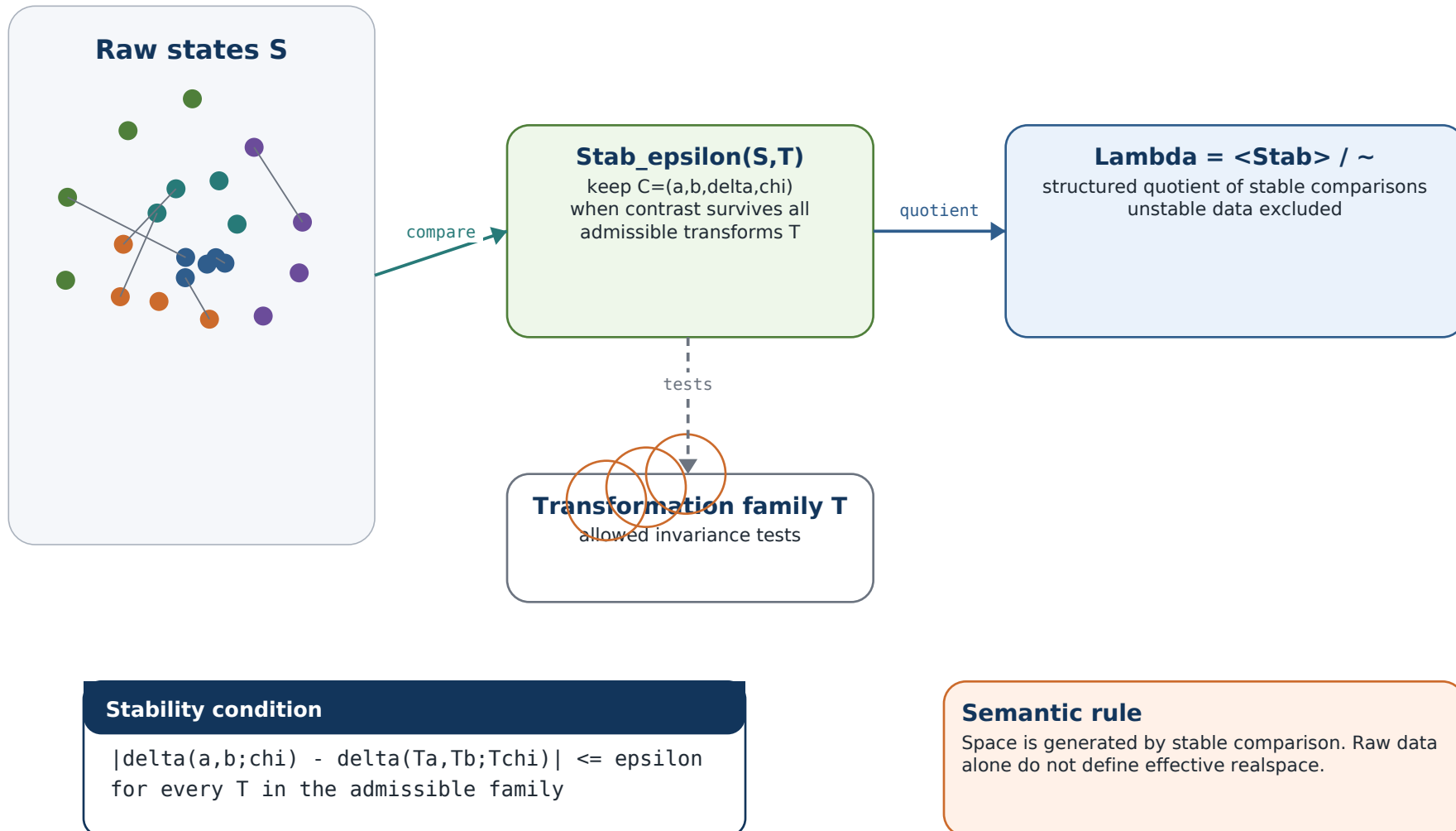
One primitive operator



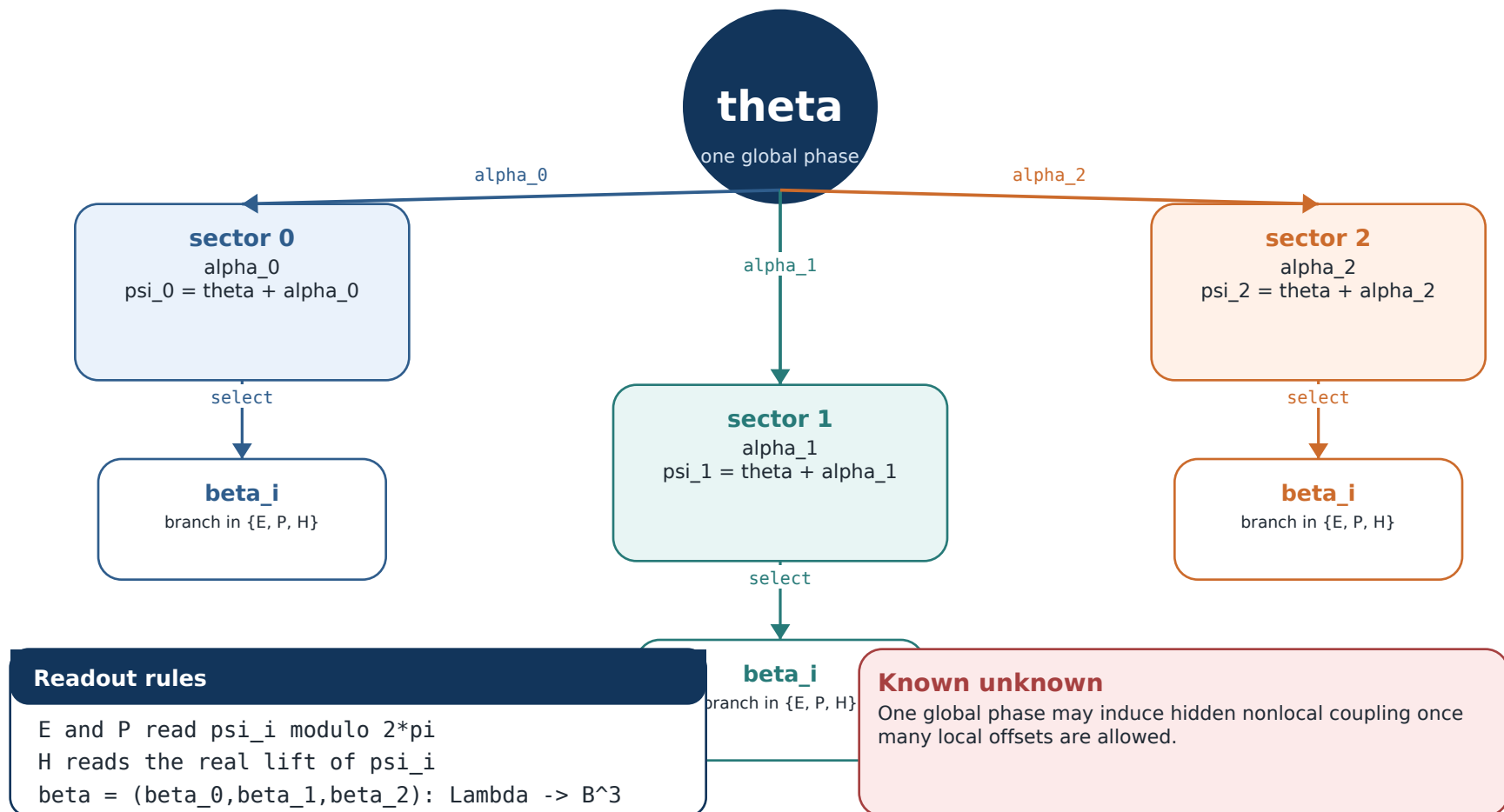
Typed dependency chain



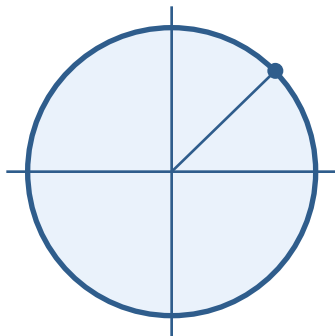
Latentization



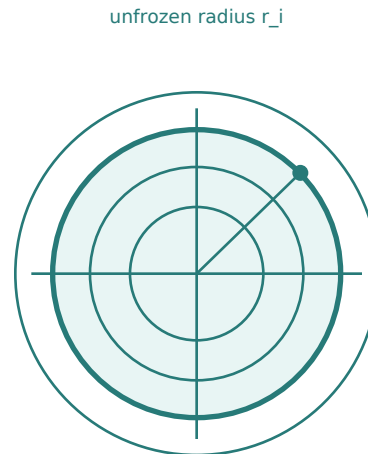
Phase coherence



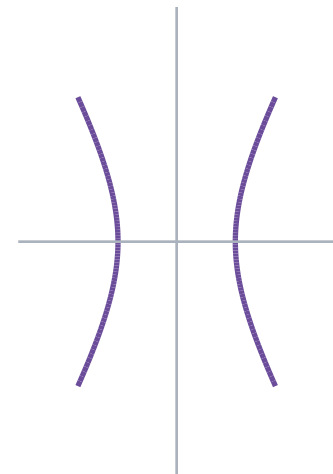
Universal transverse map



E: elliptic cylinder



P: polar parent



Compact leaf

$$\begin{aligned}\text{Gamma}_E(\rho, \psi) &= \\ &\rho(\cos \psi, \sin \psi) \\ q_E &= u^2 + v^2\end{aligned}$$

Radial parent

$$\begin{aligned}\text{Gamma}_P(\rho, \psi) &= \\ &\rho(\cos \psi, \sin \psi) \\ \rho &= r_i(\lambda)\end{aligned}$$

Signature flip

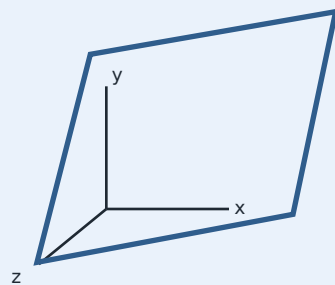
$$\begin{aligned}\text{Gamma}_H(\rho, \psi, \sigma) &= \\ &\sigma \rho(\cosh \psi, \sinh \psi) \\ q_H &= u^2 - v^2\end{aligned}$$

Shared invariant

For every branch b , $q_b(\text{Gamma}_b(\rho, \psi, \sigma)) = \rho^2$.

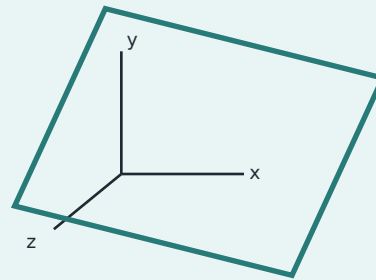
Projection geometry

$$J_0(u, v) = (\theta, u, v)$$



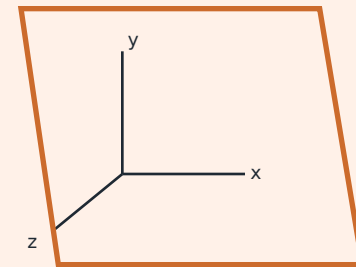
normal to x-axis

$$J_1(u, v) = (v, \theta, u)$$



normal to y-axis

$$J_2(u, v) = (u, v, \theta)$$



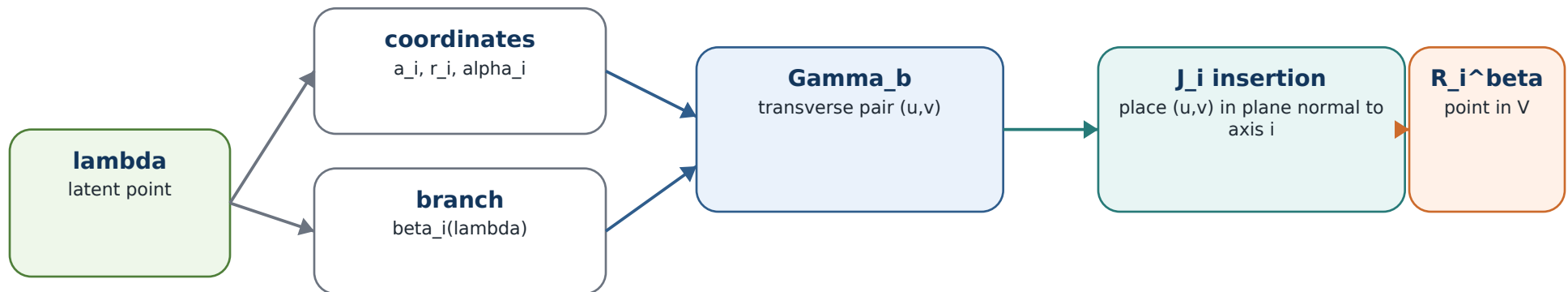
normal to z-axis

Axis-invariant construction rule

$$R_i^{\beta}(\lambda) = a_i(\lambda)e_i + J_i \Gamma_{\beta_i(\lambda)}(\rho_i^{\beta}(\lambda), \theta + \alpha_i(\lambda), \sigma_i)$$

No axis-specific branch formula remains primitive after J_i and Γ_b are defined.

R_i construction



Axis 0 example

```

i=0:
R_0^E = (a0,
  R0 cos psi0, R0 sin psi0)
R_0^H = (a0,
  s0 r0 cosh psi0,
  s0 r0 sinh psi0)
  
```

Radius and phase

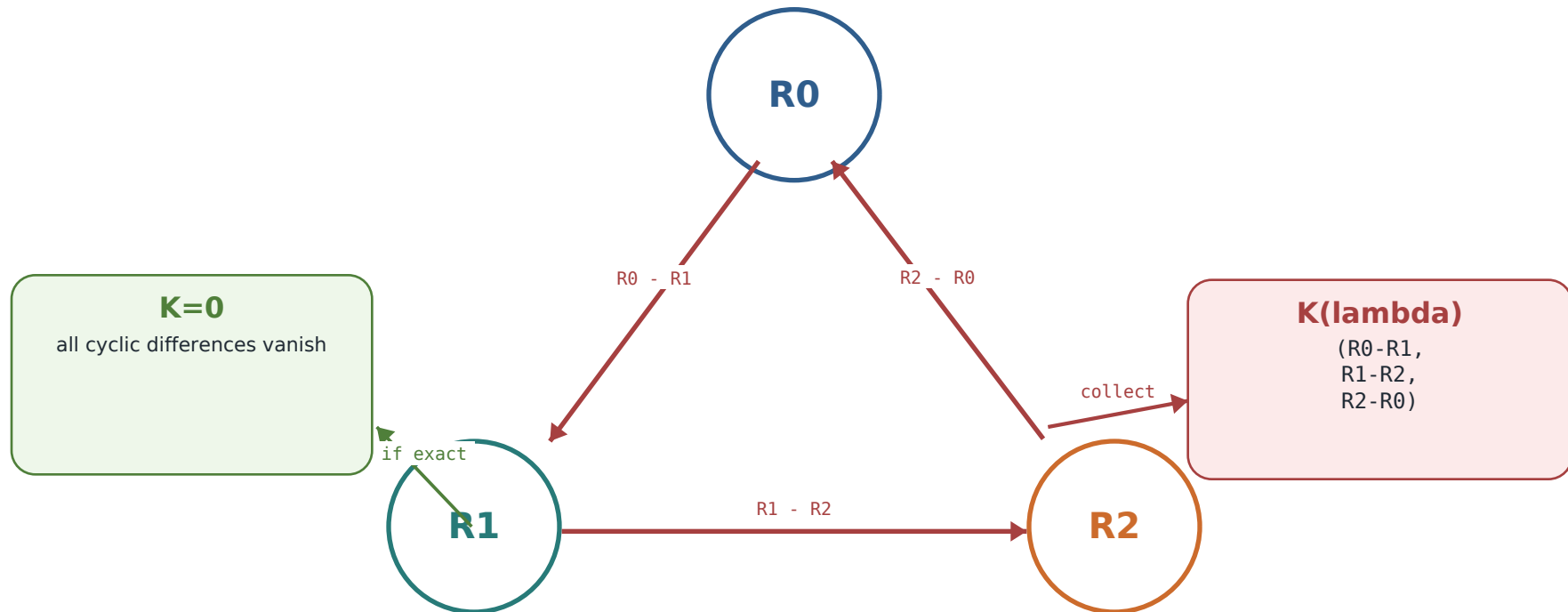
```

rho_i^b = R_i
  if b=E
rho_i^b = r_i(lambda)
  if b=P,H
psi_i = theta + alpha_i
  
```

Triple output

Full projection triple:
 $R^{\beta_{\theta}(\lambda)} = (R_0^\beta, R_1^\beta, R_2^\beta)$ in V^3

Axis coboundary



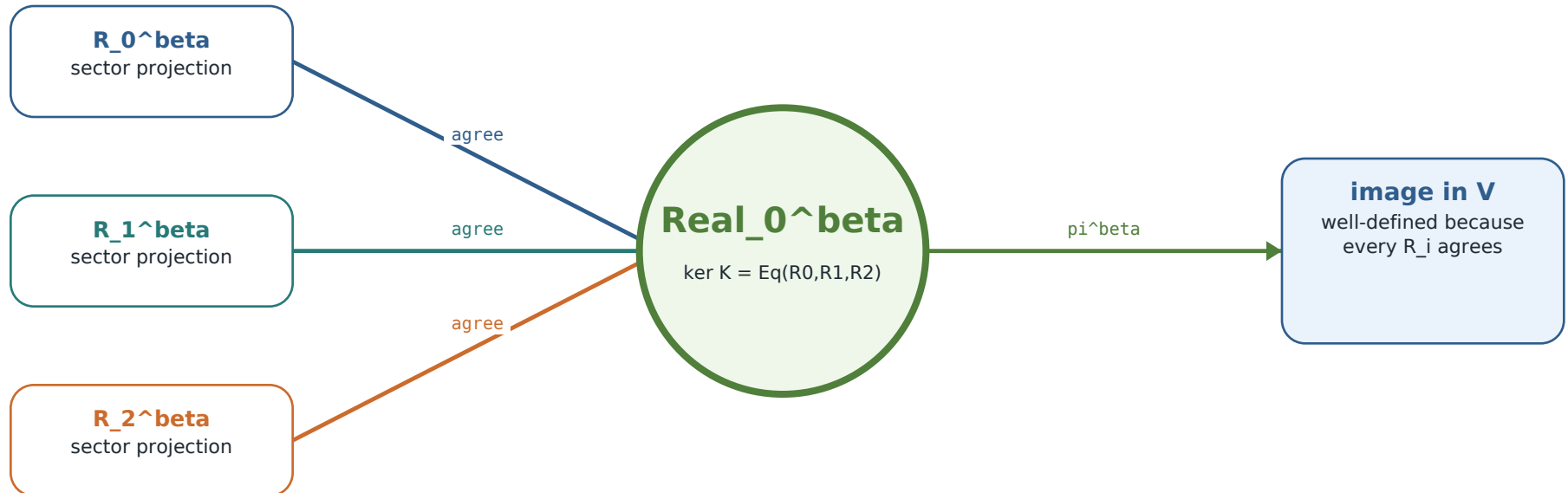
Reusable coboundary pattern

$\text{delta_A}(X_0, X_1, X_2) = (X_0 - X_1, X_1 - X_2, X_2 - X_0)$

$K_{\{\theta, \beta\}}(\lambda) = \text{delta_A}(R_0^{\beta}(\lambda), R_1^{\beta}(\lambda), R_2^{\beta}(\lambda))$

The same coboundary pattern later controls infinitesimal defect and relative curvature.

Real_0



Kernel-equalizer identity

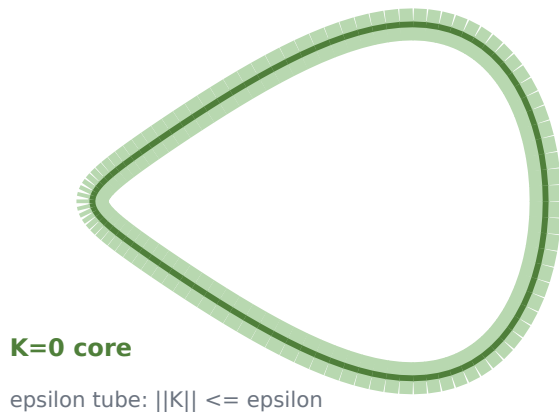
$$\begin{aligned} \text{Real}^\beta_0 &= \ker K_{\{\theta, \beta\}} \\ &= \{\lambda : K(\lambda) = 0\} \\ &= \text{Eq}(R_0^\beta, R_1^\beta, R_2^\beta) \end{aligned}$$

Realspace image projection

$$\begin{aligned} \pi^\beta(\lambda) &= R_i^\beta(\lambda) \\ \text{for any } i \text{ in } \{0, 1, 2\} \\ \text{choice is irrelevant only on } \text{Real}_0^\beta \end{aligned}$$

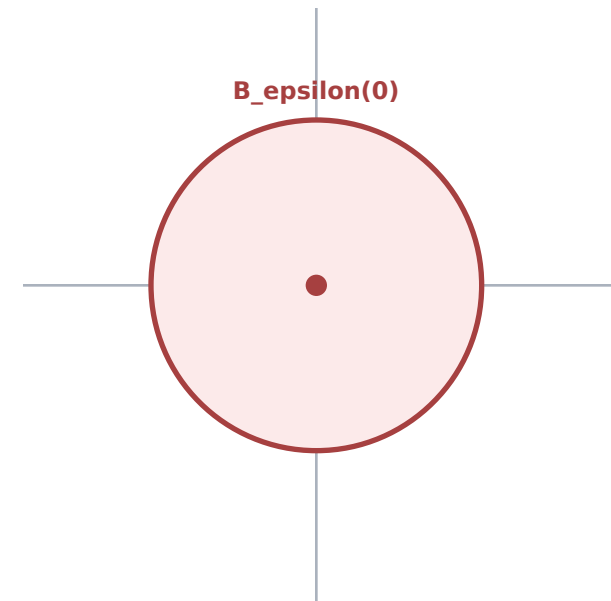
Real_epsilon

Latent comparison space Lambda



K

Defect space V^3



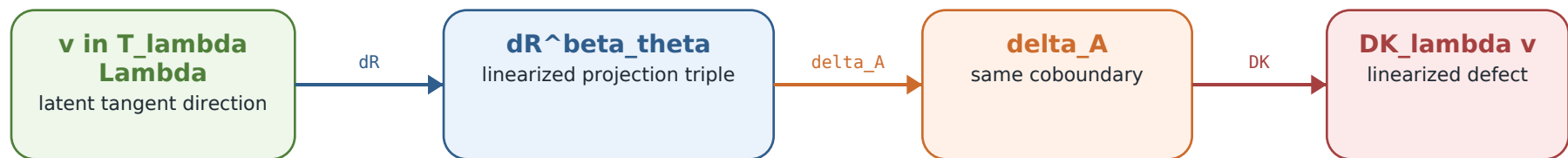
Preimage definition

$\text{Real}^{\beta_{\epsilon}} = K^{-1}_{\{\theta, \beta\}}(B_{\epsilon}(0))$
Finite realspace is compatibility up to tolerance.

Interpretation

Persistent finite defect need not be error. Under admissible minimization it can become structure.

Linearized defect



Chain rule

$$DK_{\lambda} = dK_{\{\theta, \beta\}}|_{\lambda}$$

$$DK_{\lambda} = \delta_A \circ dR^{\beta, \theta}|_{\lambda}$$

Defect-normal metric

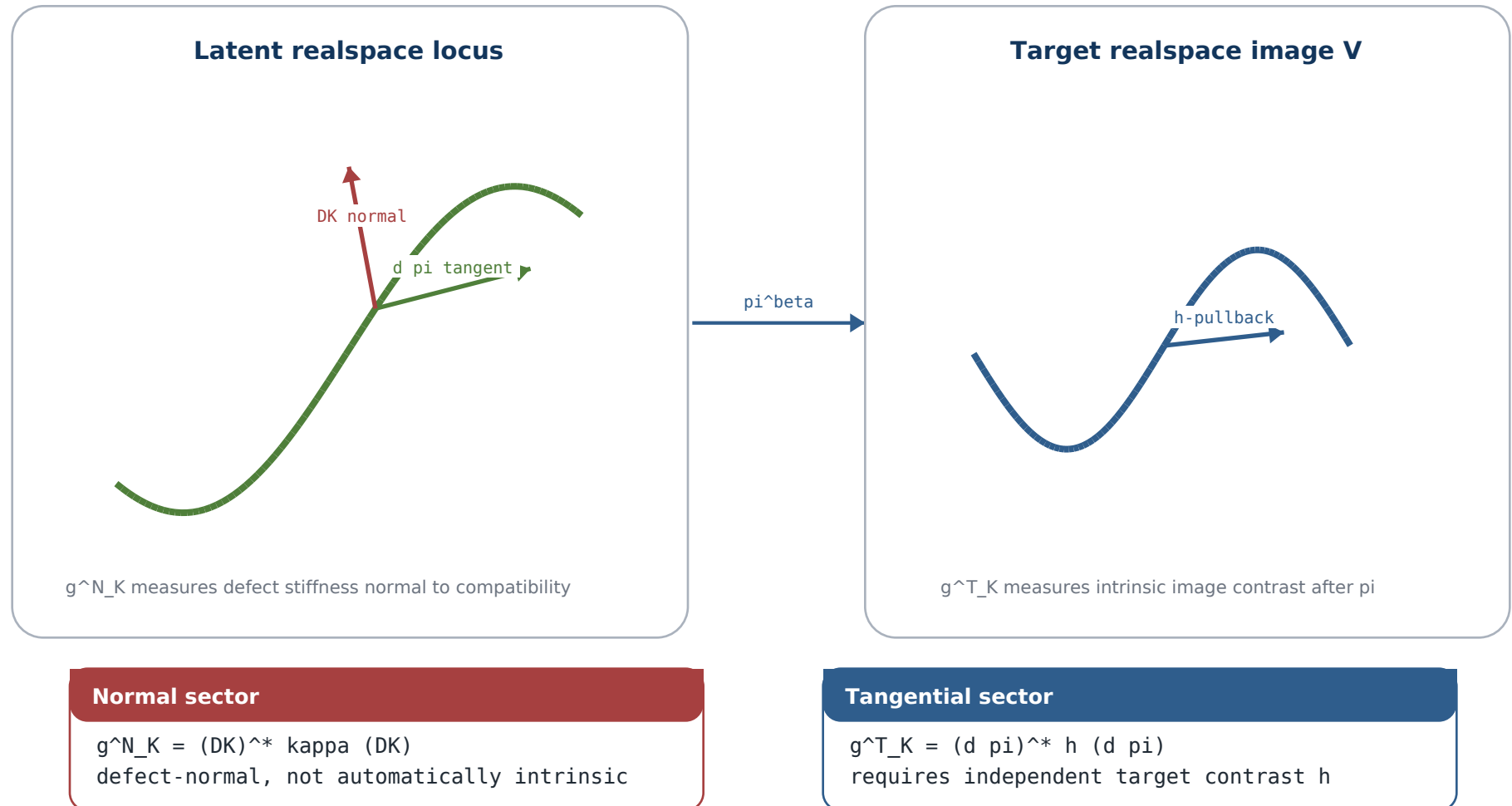
$$Q_{K, \lambda}(v) = \kappa_{\lambda}(DK v, DK v)$$

$$g^N_K = (DK)^* \kappa (DK)$$

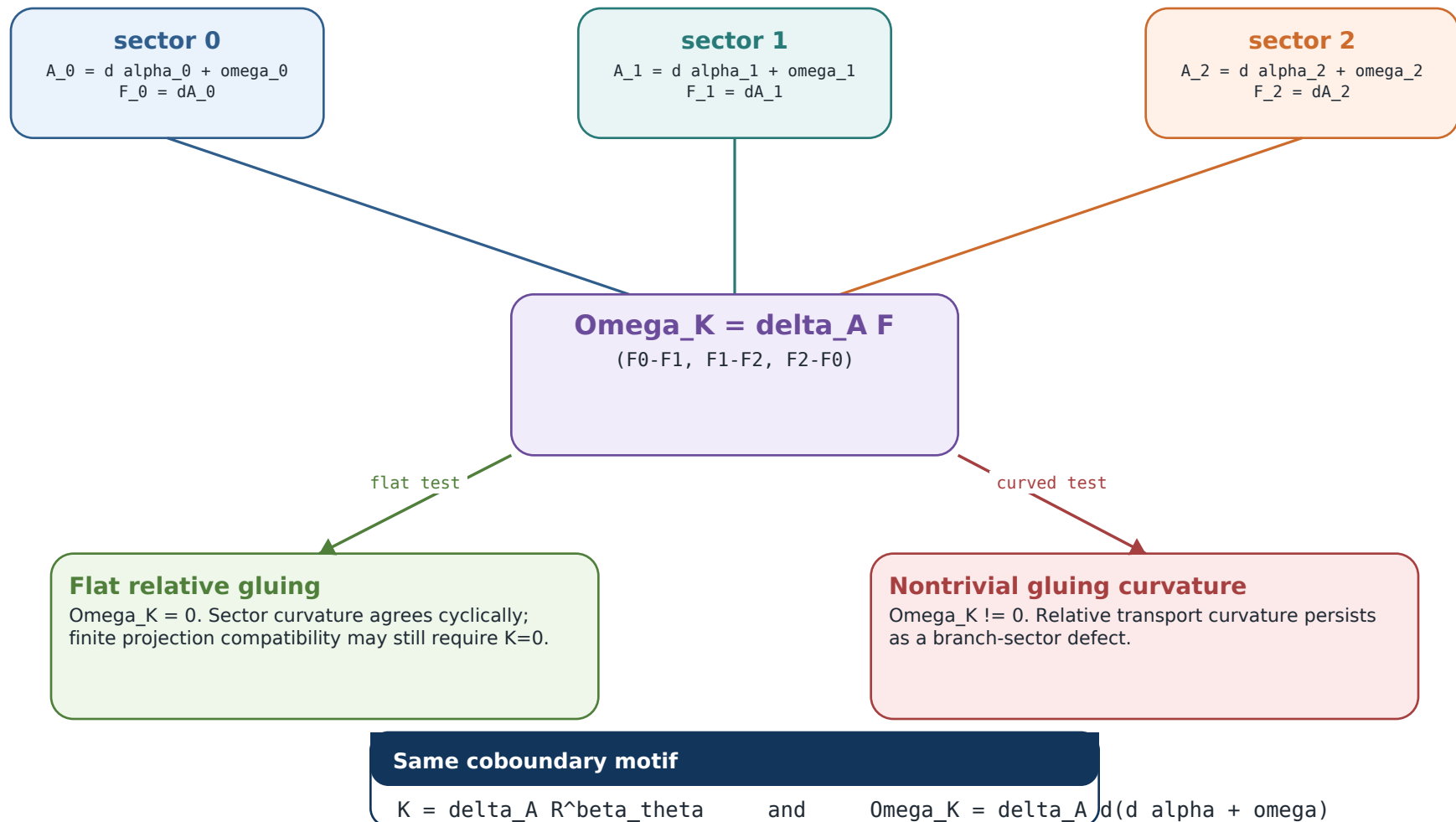
Null directions require classification

v in $\ker DK_{\lambda}$ may be gauge, physical tangent, or inadmissible degeneracy. The calculus does not decide this automatically.

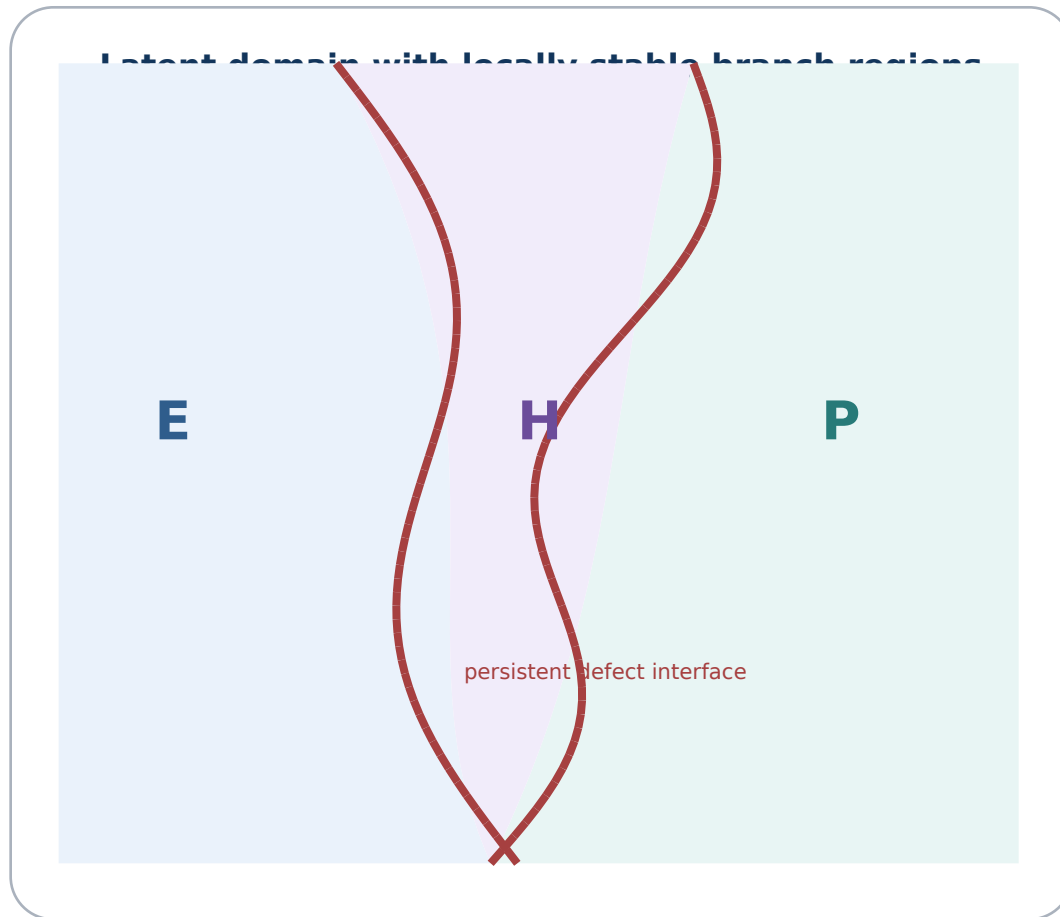
Metric separation



delta_A reused



No branch flicker



Admissibility law

```
TV(beta) < infinity  
supp(d beta) subset  
{||K|| > eta} union boundary union Sigma_prescrib
```

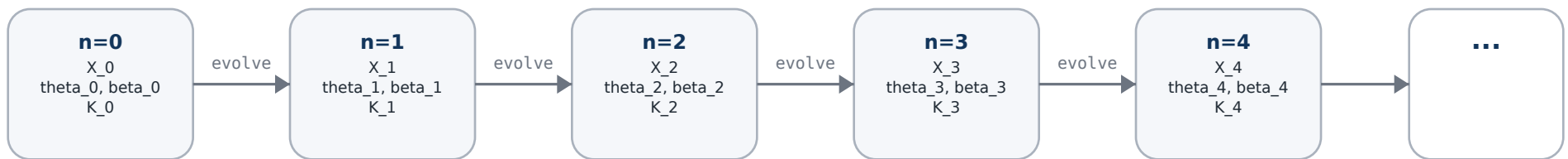
Transition cost J_{beta}

Penalizes spatial, latent, or temporal branch changes. In smooth models use $TV(\text{beta})$; in discrete models count weighted branch changes.

Known unknown

Different J_{beta} choices may produce different stable interfaces, topology changes, and defect concentrations.

History selection



Action principle

$$S[H] = \sum_n (||K_n||^2_{\kappa_n} + \lambda R_n + \mu B_n + \nu J_{\beta,n})$$

Valid history: $H^* \in \text{Crit}(S)$

Discrete update: $X_{\{n+1\}}$ locally minimizes defect + regularity + boundary + branch variation.

defect cost

$$||K||^2$$

regularity

$$R_n$$

boundary

$$B_n$$

branch variation

$$J_{\beta}$$

Persistent defect can become structure under constrained minimization.

Invariance controls

Global phase gauge

$\theta \rightarrow \theta + c$
 $\alpha_i \rightarrow \alpha_i - c$
 ψ_i invariant
 K invariant

Local connection gauge

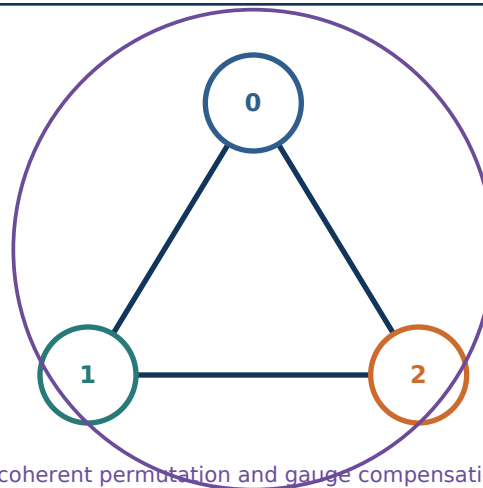
$\alpha_i \rightarrow \alpha_i + \eta_i$
 $\omega_i \rightarrow \omega_i - d \eta_i$
 A_i preserved
 F_i, Ω_K invariant

Axis permutation S3

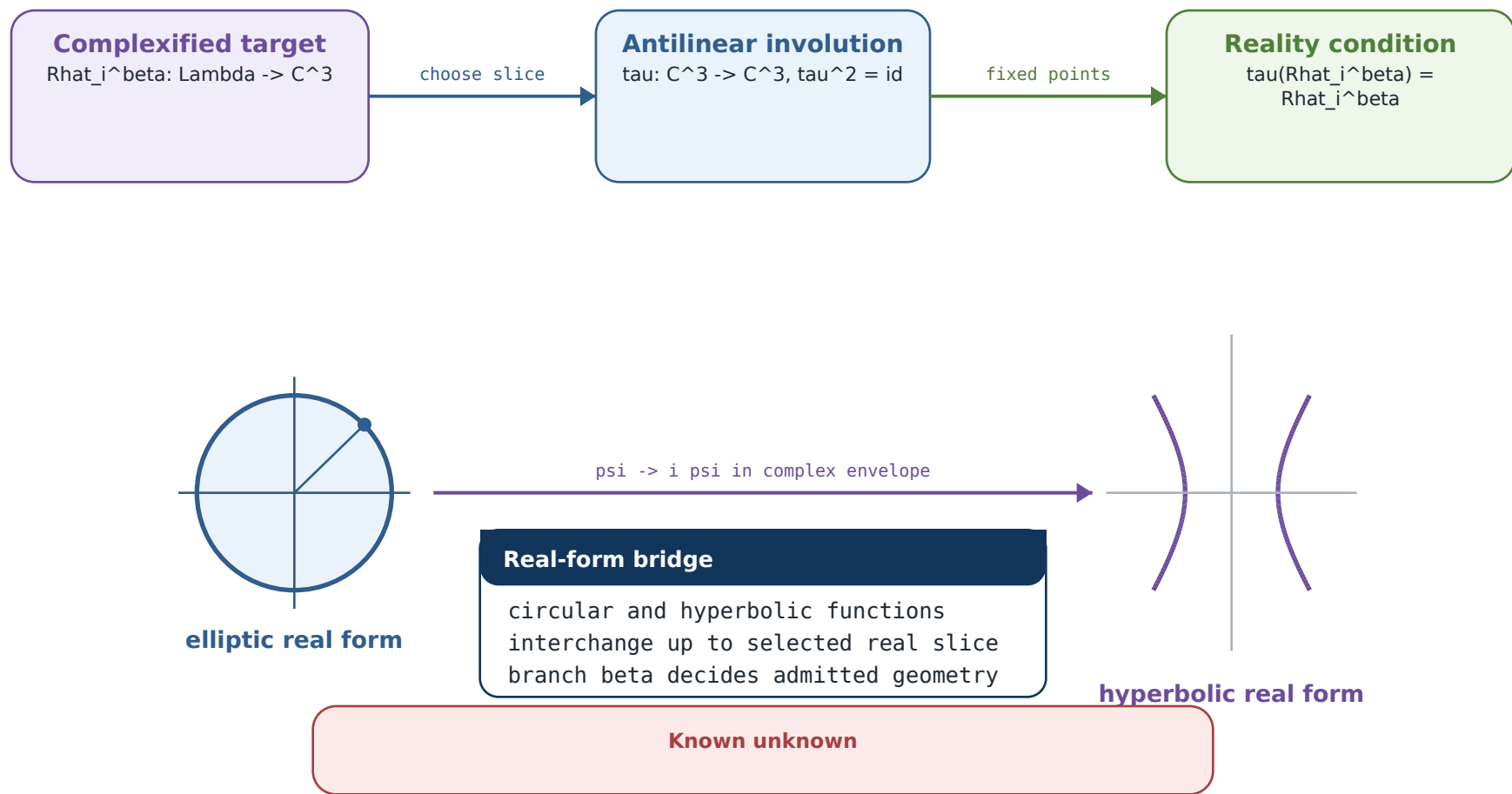
permute axes coherently
 $K_{\{\theta, \pi \beta\}}(\pi \lambda)$
 $= \pi K_{\{\theta, \beta\}}(\lambda)$
Real_epsilon transforms

Natural symmetry envelope

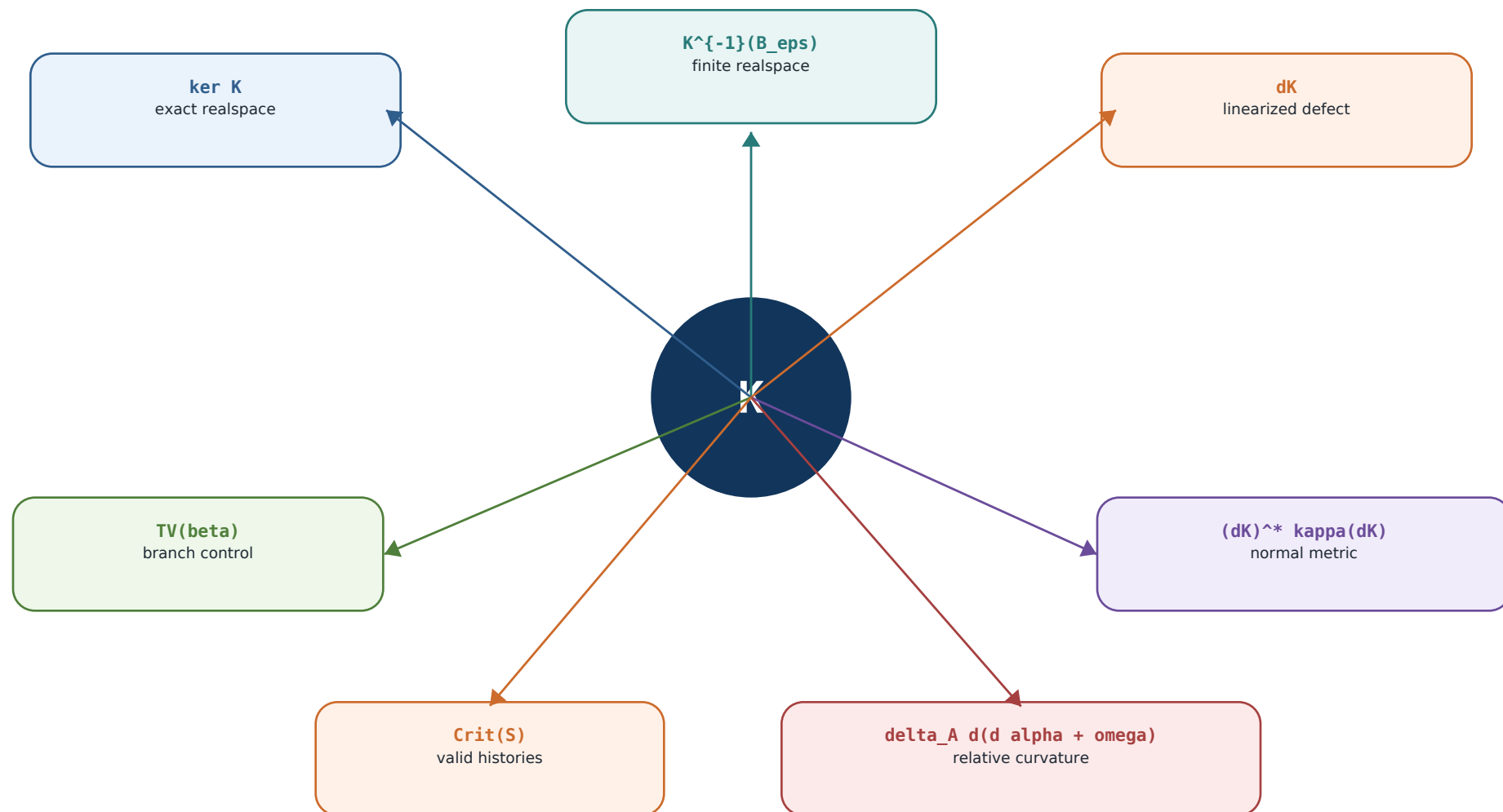
$S_3 \text{ semidirect } G_\theta \text{ semidirect } G_A$



Real-form selection



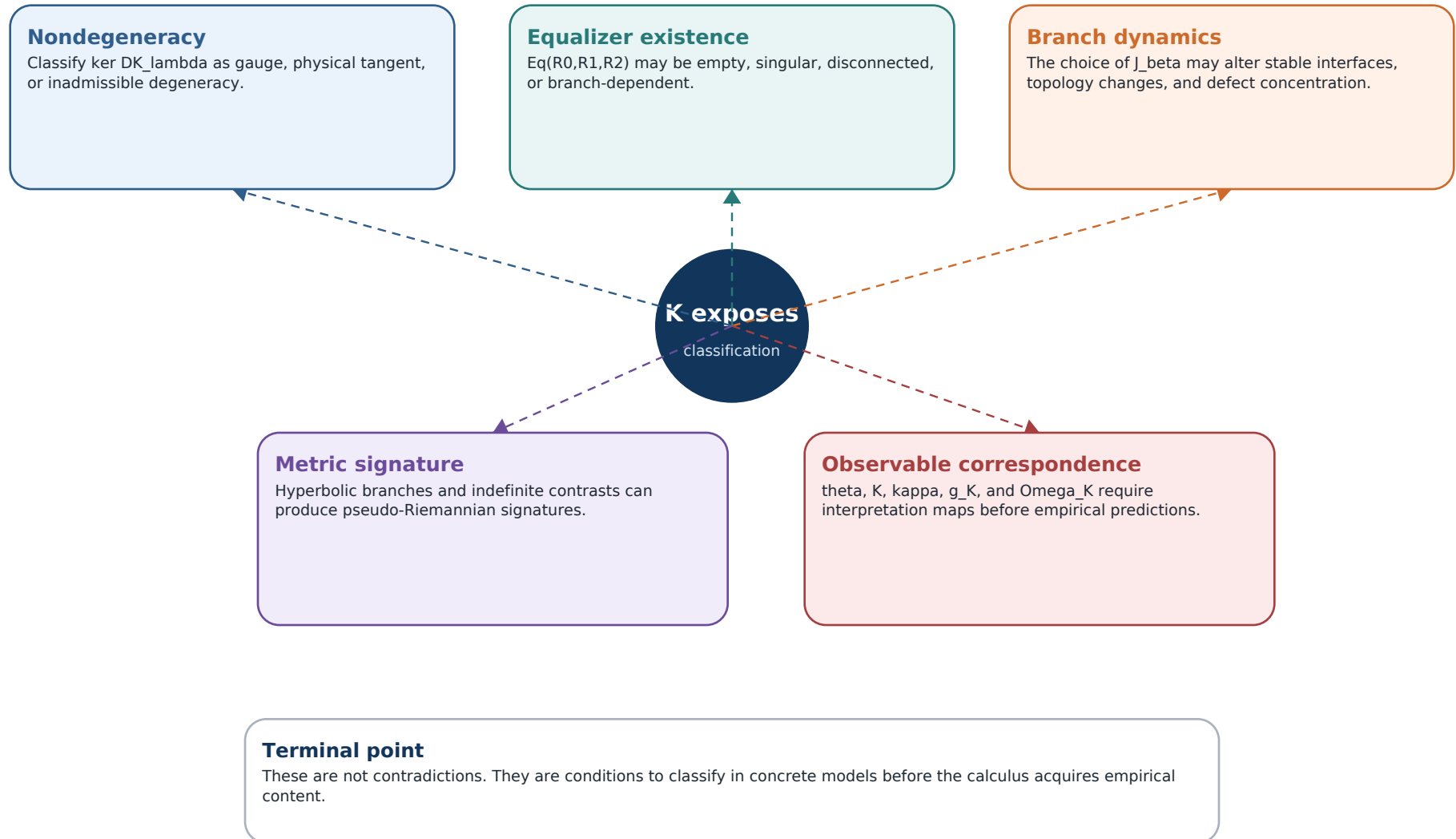
Formal closure



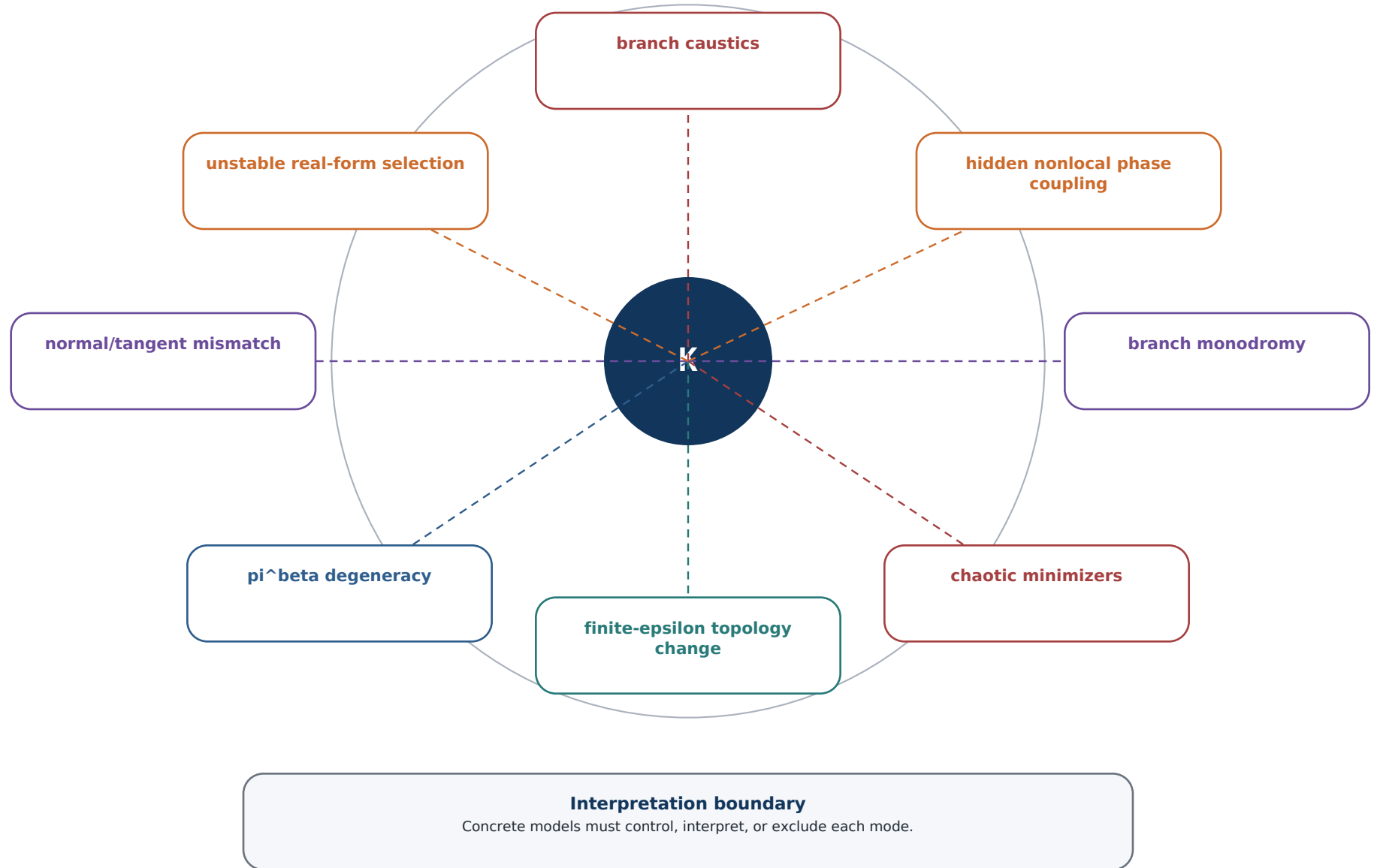
Closure signature

UCK = < K ; ker, preimage ball, d, pullback, δ_A , TV, Crit >

Classification tasks



Study targets



UCK model constraints



K approximately 0

projection defect controlled



Omega_K controlled

relative gluing curvature bounded or flat



TV(beta) finite

branch field admissible; no arbitrary flicker



H in Crit(S)

history selected by stationarity or local minimization



metric sector nondegenerate

normal and/or tangential metric classified

Terminal statement

$UCK = (L_{\epsilon}, \theta, B, \Gamma, J, R, \delta_A, K, \ker, dK, \Omega_K, \kappa, g^N_K, g^T_K, S)$
primitive: $K_{\{\theta, \beta\}} = \delta_A \circ R^{\beta}_{\theta}$